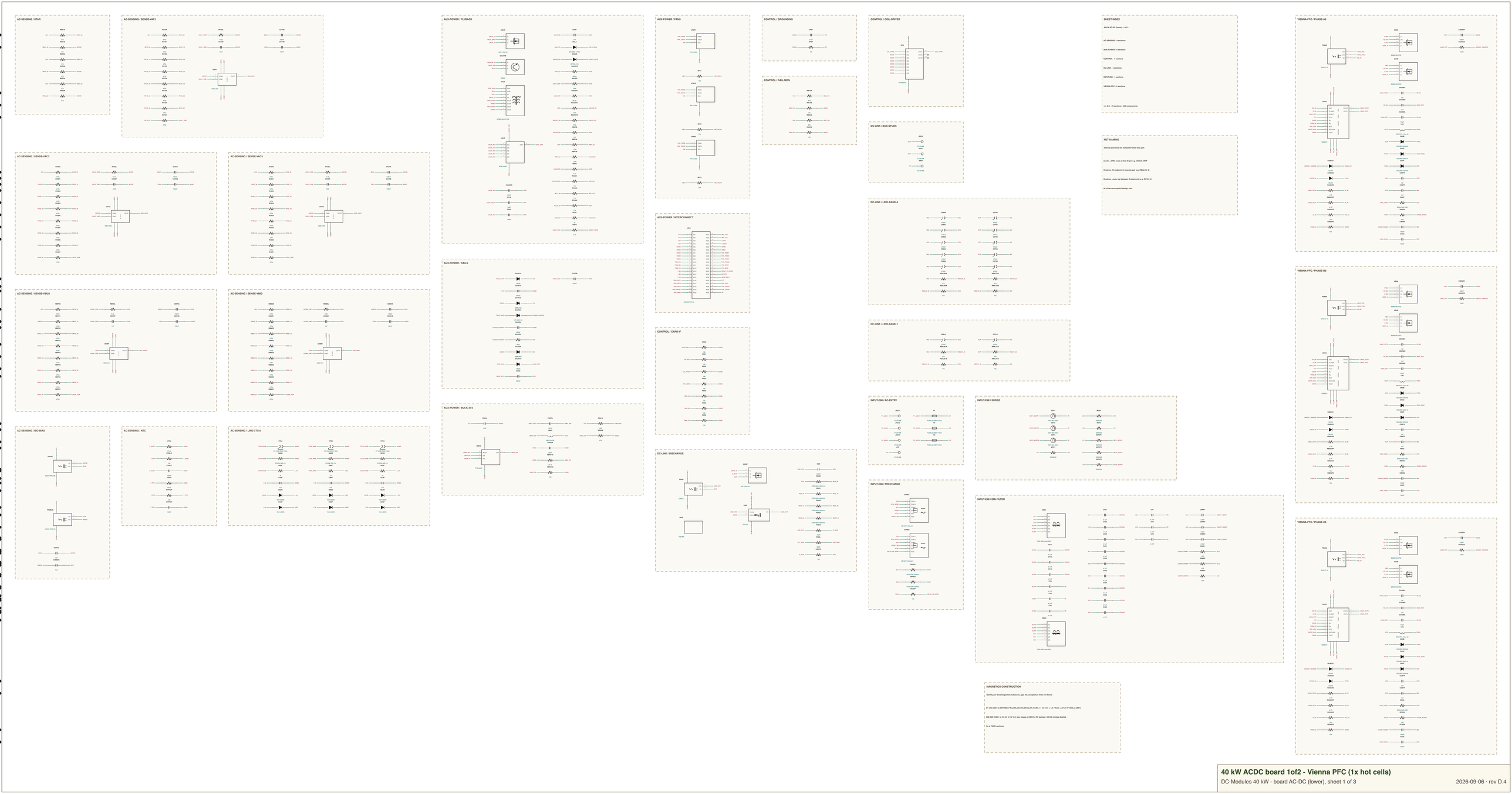

AC-SENSING / STAR
AC-SENSING / SENSE-VAC1
AC-SENSING / SENSE-VAC2
AC-SENSING / SENSE-VAC3
AC-SENSING / SENSE-VBUS
AC-SENSING / SENSE-VMID
AC-SENSING / ISO-BIAS
AC-SENSING / NTC
AC-SENSING / LINE-CTS-0
AUX-POWER / FLYBACK
AUX-POWER / FANS
AUX-POWER / INTERCONNECT
AUX-POWER / RAILS
AUX-POWER / BUCK-3V3
CONTROL / GROUNDING
CONTROL / COIL-DRIVER
CONTROL / RAIL-MON
CONTROL / CARD-IF
DC-LINK / DISCHARGE
DC-LINK / BUS-STUDS
DC-LINK / LINK-BANK-0
DC-LINK / LINK-BANK-1
INPUT-EMI / AC-ENTRY
INPUT-EMI / SURGE
INPUT-EMI / PRECHARGE
INPUT-EMI / EMI-FILTER
SHEET INDEX
40 kW AC-DC (lower) · 1 of 3
AC-SENSING · 9 sections
AUX-POWER · 5 sections
CONTROL · 4 sections
DC-LINK · 4 sections
INPUT-EMI · 4 sections
VIENNA-PFC · 3 sections
~ rev D.0 · 29 sections · 320 components
NET NAMING
Internal junctions are named for what they join
Grands, «PIN» rests at that IC pin e.g. U3V3A, GNF
Routers, M-mid/point of a series pair e.g. R6A/7K, 3K
Rupture, «into» top between Rrudimentu e.g. RV10, 21
all «Vars are explicit design nets
VIENNA-PFC / PHASE-A0
VIENNA-PFC / PHASE-B0
VIENNA-PFC / PHASE-C0
MAGNETICS CONSTRUCTION
Identity per dimachnagnetics and (turns, gap, Ihs, acceptance times Ave there)
D1 LAA-L21: 5x SOTTMAKZ Kuelite (CATALOG ALOT), N=26 +/- 1 turn into ~ L2: 110uH, +/-07uH @ 0.5mA pk (D1!)
EM EMI: CMC1 + 10x X2 4.7nF to 1 star stages + CMC2 + RC damper; D6 DM choices deleted
D or TWIRE sections
40 kW ACDC board 1of2 - Vienna PFC (1x hot cells)
DC-Modules 40 kW - board AC-DC (lower), sheet 1 of 3
2026-09-06 · rev D.4